The UAT Lagrangian: Derivation of the Quantum Braking Action and Closure of the Causal Coherence Parameters

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Abstract

We present the Lagrangian formulation of the Unified Applicable Time (UAT) framework, a scalar-tensor theory in which a causal field ϕ couples non-minimally to gravity and governs the local flow of time. The action S_{UAT} contains three parameters: the vacuum scale η, the non-minimal coupling ξ, and the self-coupling λ. We show that these parameters are not free but are rigidly fixed by three independent constraints derived from the UAT/UCP architecture: (i) $\eta \equiv \kappa_{\mathrm{crit}} = 4.978$, the Ivancho causal limit; (ii) $\phi_* = 0.07\,\eta$, the 7% thermal calibration margin at recombination; and (iii) the quantum braking factor $k_{\mathrm{early}} = 0.967$, which determines ξ. From these we obtain $\xi = -0.2810$ and $\lambda \approx 3.08 \times 10^{-112}$ in Planck units. The resulting effective Friedmann equation reproduces the UAT expansion history and embeds the Hartman saturation and the causal coherence limit within a single fundamental action. This work completes the formal mathematical foundation of the UAT/UCP programme.

1 Introduction

The Unified Applicable Time (UAT) framework [1] and its corollary, the Unified Causal Principle (UCP) [2], propose that the macroscopic flow of time is an emergent phenomenon regulated by a fundamental causal coherence constant $\kappa_{\mathrm{crit}} \approx 4.978$. In the cosmological setting, this regulation manifests as a "quantum brake" —the parameter $k_{\mathrm{early}} \approx 0.967$— that modifies the early-universe expansion rate and resolves the Hubble tension [3].

Previous phenomenological analyses have demonstrated that UAT is not ruled out by current SNe, CMB and BAO data; it performs at least as well as ΛCDM [4]. What has been missing is a rigorous derivation of the UAT dynamics from an action principle. This article fills that gap.

2 The UAT Action

We postulate that the "applicable time" is encoded in a real scalar field ϕ whose value controls the local rate of temporal flow. The action, in natural units ($\hbar = c = 1$) but with gravitational coupling explicitly retained, reads

$$S_{\mathrm{UAT}} = \int d^4x\sqrt{-g}\left[\frac{M_{\mathrm{Pl}}^2}{2}R - \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi) - \frac{\xi}{2}R\phi^2 + \mathcal{L}_m\right], \tag{1}$$

where $M_{\mathrm{Pl}} = (8\pi G)^{-1/2}$ is the reduced Planck mass, R the Ricci scalar, and $V(\phi)$ a double-well potential

$$V(\phi) = \frac{\lambda}{4}(\phi^2 - \eta^2)^2 + V_0, \tag{2}$$

with V_0 tuned to match the present-day cosmological constant ($V_0 \approx 2.5 \times 10^{-122}\, M_{\mathrm{Pl}}^4$).

3 Equations of Motion and the Quantum Brake

In a flat FLRW background, the field equation and the Friedmann equation become

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} + \xi R\phi = 0, \tag{3}$$

$$3M_{\mathrm{Pl}}^2 H^2 = \rho_m + \rho_r + \rho_\phi, \tag{4}$$

with $\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi) + 3\xi H^2\phi^2 + 6\xi H\phi\dot{\phi}$. In the slow-roll regime of the early universe, $\rho_\phi \approx 3\xi H^2\phi^2$, and the Friedmann equation reduces to

$$H^2 = \frac{1}{3M_{\mathrm{Pl}}^2}\frac{\rho_m + \rho_r}{1 - \xi\phi^2}. \tag{5}$$

Comparing with the phenomenological UAT law $H_{\mathrm{UAT}}^2 = (3M_{\mathrm{Pl}}^2)^{-1}k_{\mathrm{early}}(\rho_m + \rho_r)$ yields the fundamental relation

$$k_{\mathrm{early}} = \frac{1}{1 - \xi\phi_*^2}, \tag{6}$$

where ϕ_* is the value of the scalar field at recombination.

4 Closure of Parameters

The UAT/UCP framework provides three independent, non-adjustable constraints.

4.1 Vacuum scale from the causal limit

The maximum excursion of the field before causal coherence is lost is set by the Ivancho constant:

$$\boxed{\eta \equiv \kappa_{\mathrm{crit}} = 4.978 \quad \text{(in Planck-reduced units)}}. \tag{7}$$

4.2 Thermal calibration at recombination

The 7% thermal noise margin fixes the residual field at recombination:

$$\phi_* = 0.07\,\eta = 0.07 \times 4.978 = 0.34846. \tag{8}$$

4.3 Non-minimal coupling from the quantum brake

Inserting ϕ_* and $k_{\text{early}} = 0.967$ into (6) gives

$$\xi = \frac{1 - 1/k_{\text{early}}}{\phi_*^2} = \frac{-0.0341}{0.1214} \approx -0.2810. \tag{9}$$

The negative sign confirms the repulsive scalar pressure that opposes gravitational collapse.

4.4 Self-coupling from the recombination density

The effective potential $V_{\text{ef}}(\phi) = V(\phi) + \frac{\xi}{2} R\phi^2$ becomes unstable when $V''_{\text{ef}}(\eta) = 0$, i.e. when

$$2\lambda\eta^2 + \xi R_c = 0. \tag{10}$$

At recombination, $R_c \approx 8\pi G\,\rho_{\text{rec}}$ with $\rho_{\text{rec}} \approx (1.1\text{ eV})^4 \approx 2.17 \times 10^{-111}\, M_{\text{Pl}}^4$. Solving for λ yields

$$\boxed{\lambda = -\frac{\xi R_c}{2\eta^2} \approx 3.08 \times 10^{-112}}. \tag{11}$$

5 Effective Friedmann Equation and Predictions

With the parameters fixed, the effective Friedmann equation is

$$H^2 = \frac{1}{3M_{\text{Pl}}^2}\, k_{\text{early}}(\rho_m + \rho_r), \qquad k_{\text{early}} = 0.967. \tag{12}$$

This reproduces the UAT expansion history that has been tested against Pantheon+, CMB and BAO data [4]. The companion document `UAT_Observational_Predictions.pdf` provides the predicted shifts of the acoustic peaks relative to ΛCDM.

6 Conclusion

We have derived the Lagrangian of the UAT framework and demonstrated that its three parameters are rigidly determined by the fundamental UAT/UCP constants. The resulting action anchors the quantum brake, the thermodynamic overdrive threshold, and the causal coherence limit within a single scalar-tensor theory. This work completes the formal mathematical foundation of the UAT/UCP programme.

References

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